Representation of data

Line graph, Bar graph Pie chart,
Histogram & Frequency Polygon

Methods of Graphical Representation of Data

- Ø In statistics, the data can be presented graphically using many methods.
- Ø Important graphical representation methods are given below:

(1). **Line Diagram**

(2). **Bar Diagram**

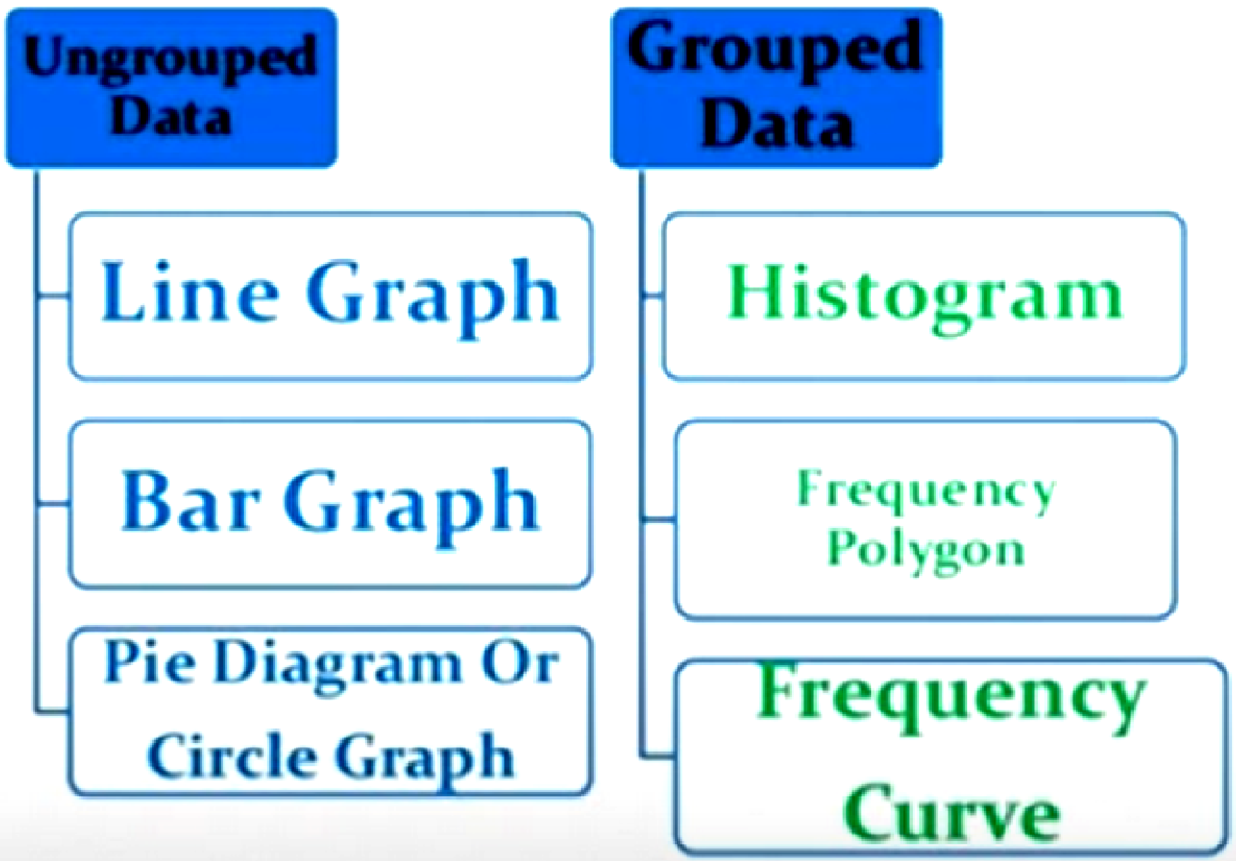
(3). **Histogram**

(4). **Frequency Polygon**

(5). **Frequency Curve**

(6). **Pie Chart (Circle Diagram)**

TYPES OF GRAPHICAL REPRESENTATION



What is bar graph?

I. Bar graph is the simplest way to represent a data.
II. In consists of rectangular bars of equal width.
III. The space between the two consecutive bars must be the same.
IV. Bars can be marked both vertically and horizontally but normally we use vertical bars.

How to Construct a Bar Graph?

Steps in construction of bar graphs/column graph:

- On a graph, draw two lines perpendicular to each other.
- The horizontal line is **x-axis** and vertical line is **y-axis**
- Along the horizontal axis, choose the uniform width of bars and uniform gap between the bars and write the names of the data items whose values are to be marked.

- Along the vertical axis, choose a suitable scale in order to determine the heights of the bars for the given values. (Frequency is taken along y-axis).

- Calculate the heights of the bars according to the scale chosen and draw the bars.

Question

150 students of class VI have popular school subjects as given below

Subject	French	English	Maths	Geography	Science
Number of Students	30	20	26	38	34

Draw the column graph/bar graph representing the above data.

Solution:

Take the subjects along **x-axis**, and the number of students along **y-axis**

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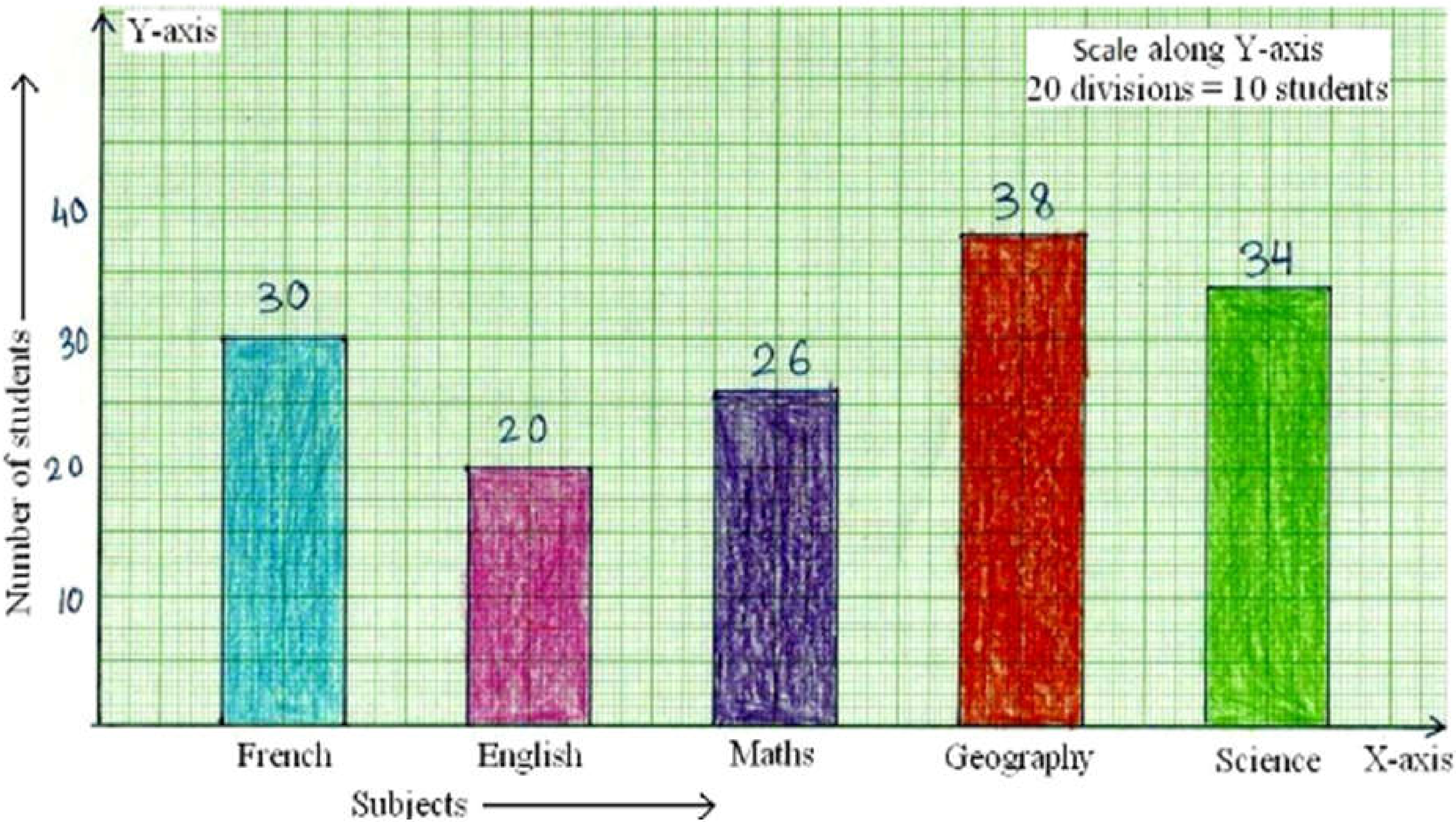


What is a Pie Chart?

A **pie chart** is a type of graph that represents the data in the circular graph

Formula

It contains different segments and sectors in which each segment and sector of a pie chart forms a specific portion of the total (percentage). The sum of all the data is equal to 360°.

We will take a full circle of 360° and follow the calculations below:

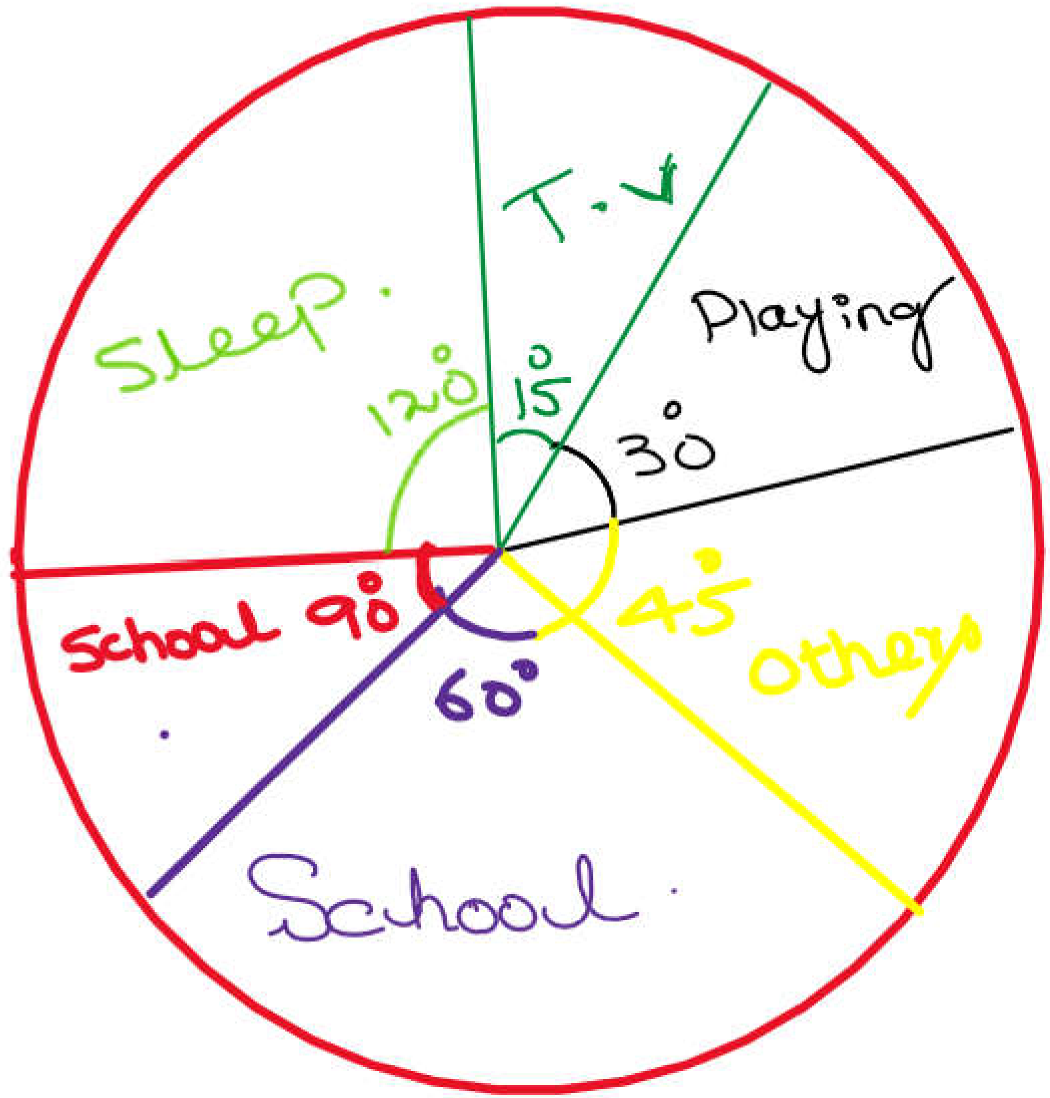
The central angle of each component

= (Value of each component/sum of values of all the components) $\times$360°

The following table shows the numbers of hours spent by a child on different events on a working day. Represent the adjoining information on a pie chart.

School	Sleep	Playing	Study	T.V	Others
6	8	2	4	1	3

Activity	No. of Hours	Measure of central angle
School	6	$(\frac{6}{24} \times 360)^\circ = 90^\circ$
Sleep	8	$(\frac{8}{24} \times 360)^\circ = 120^\circ$
Playing	2	$(\frac{2}{24} \times 360)^\circ = 30^\circ$
Study	4	$(\frac{4}{24} \times 360)^\circ = 60^\circ$
T. V.	1	$(\frac{1}{24} \times 360)^\circ = 15^\circ$
Others	3	$(\frac{3}{24} \times 360)^\circ = 45^\circ$


T.V
Sleep.
Playing
120°
15°
30°
School 90°
45°
Others
60°
School.

Histogram

A two dimensional graphical representation of a continuous frequency distribution is called a **histogram**

In histogram, the bars are placed continuously side by side with no gap between adjacent bars. That is, in histogram rectangles are erected on the class intervals of the distribution. The areas of rectangle are proportional to

Frequency Polygon

A frequency polygon is a many sided closed figure. It is constructed by plotting the class frequencies against their corresponding class marks (mid-points) and then joining the resulting points by means of straight lines. It can also be obtained by joining the mid-points of the tops of rectangles in the histograms.

Method

- Draw X-axis and Y-axis.
- Take class marks on X-axis and frequencies on Y-axis.
- Join the points by means of straight lines. The resulting figure is the required frequency polygon.

Draw a histogram and frequency polygon of given data

12,14,14,14,16,18,20,20,21,23,27,27,27,29,31,31,32,32,34,36,40,40,40,40,40,42, 51,56, 60, 65

To make a histogram and frequency polygon by hand, we must first find the **frequency distribution.**

Step 1: Calculate the range of the data set

The range is the difference between the largest value and the smallest value. We need this to figure out how much "space" we need to divide into groups. In this example:

$$\textbf{Range} = 65\text{-}21 = 53$$

Step 2: Divide the range by the number of groups you want and then round up

for number of groups

$$\sqrt{n} = \sqrt{30} = 5.4 \approx 6$$

$$\text{Class width} = \frac{53}{6} = 8.8$$

We will round this up to 9 just because it is easier to work with that way Class width = 9

Step 3: Use the class width to create your groups

We are going to start at the smallest number we have, which is 12, and count by 9 until we have 6 groups

- For example, first group will be **12 - 21** since. 12+9 =21
- Next group will be **21 - 30** since 21+9=30 and so on.
- We'll put these in a table and label them "classes". We will also add "frequency" to the table

Classes	Frequency
12 – 21	
21 – 30	
30 – 39	
39 – 48	
48 – 57	
57 – 66	

Step 4: Find the frequency for each group

We are going to count how many points are in each group. Let's start with our first group: 12 – 21. We want to count how many points are between 12 and 21 **not including 21**. **The right hand endpoint of any group isn't included in that group. It goes in the next group.** That means 21 would be in the second group and any 30 we have would be counted in the third group.

Continuing with this pattern (each group is a different color!):

	8	6	6		6	
12	18	27	31	40	42	
14	20	27	32	40	51	2
14	20	27	32	40	56	
14	21	29	34	40	60	2
16	23	31	36	40	65	

Classes Frequency

12 – 21 8

21 – 30 6

30 – 39 6

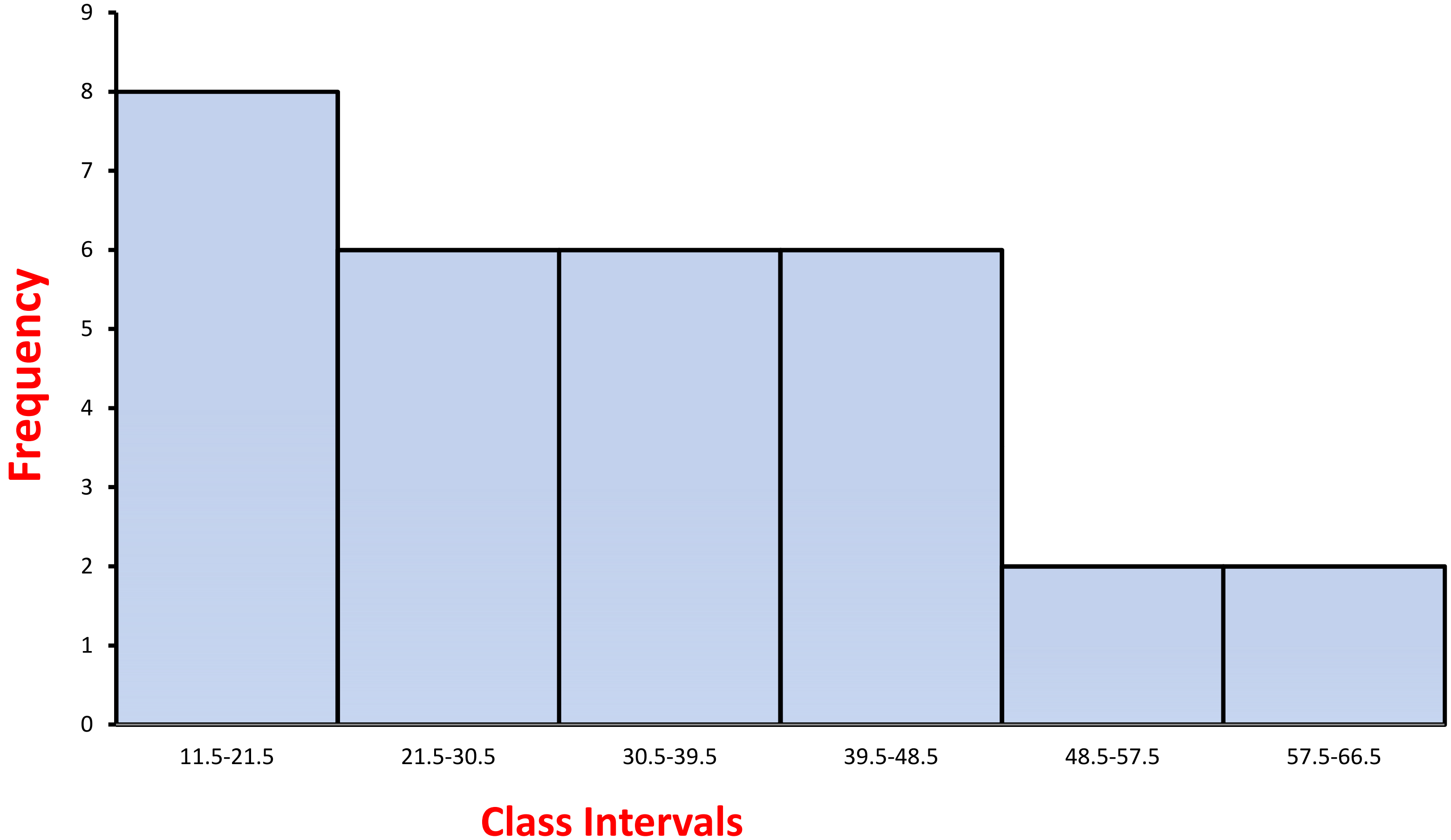
39 – 48 6

48 – 57 2

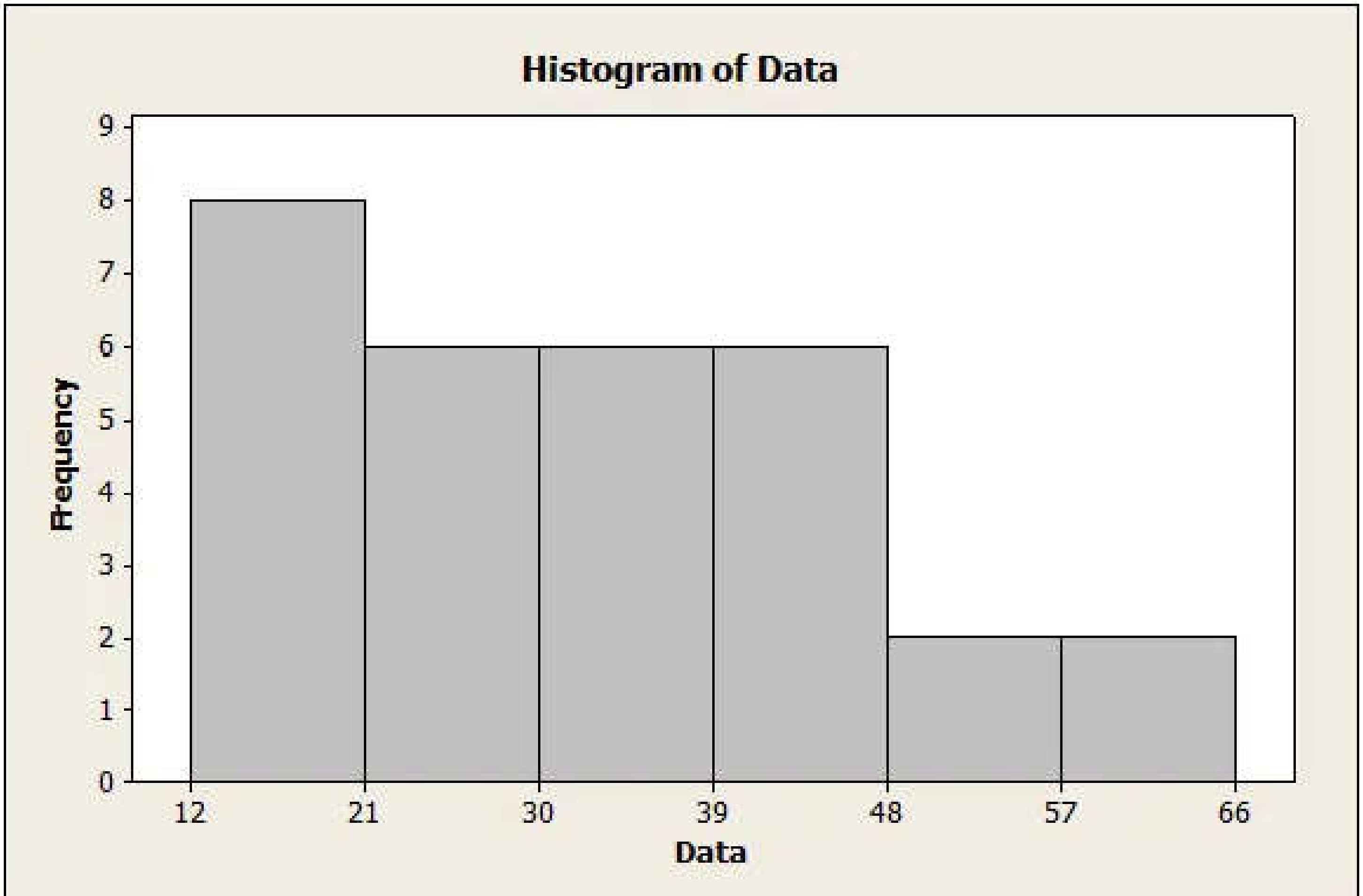
57 – 66 2

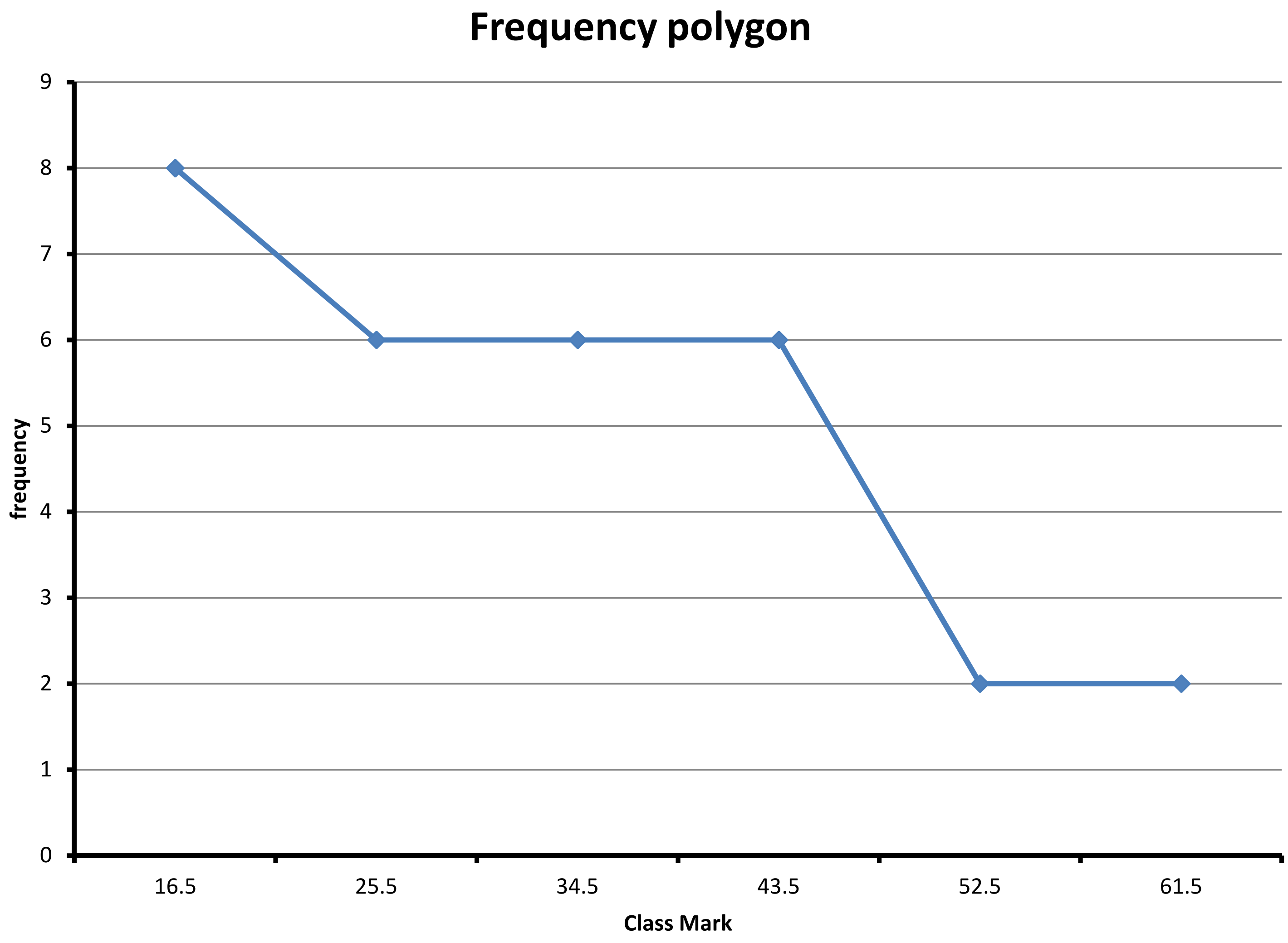
Class intervals	Frequency	Class mark	Class boundaries
12-21	8	16.5	11.5-21.5
21-30	6	25.5	21.5-30.5
30-39	6	34.5	30.5-39.5
39-48	6	43.5	39.5-48.5
48-57	2	52.5	48.5-57.5
57-66	2	61.5	57.5-66.5

Histogram with class boundaries



Another way to draw a histogram




Frequency polygon
frequency
9
8
7
6
5
4
3
2
1
0
16.5
25.5
34.5
43.5
52.5
61.5
Class Mark

Histogram and Frequency polygon

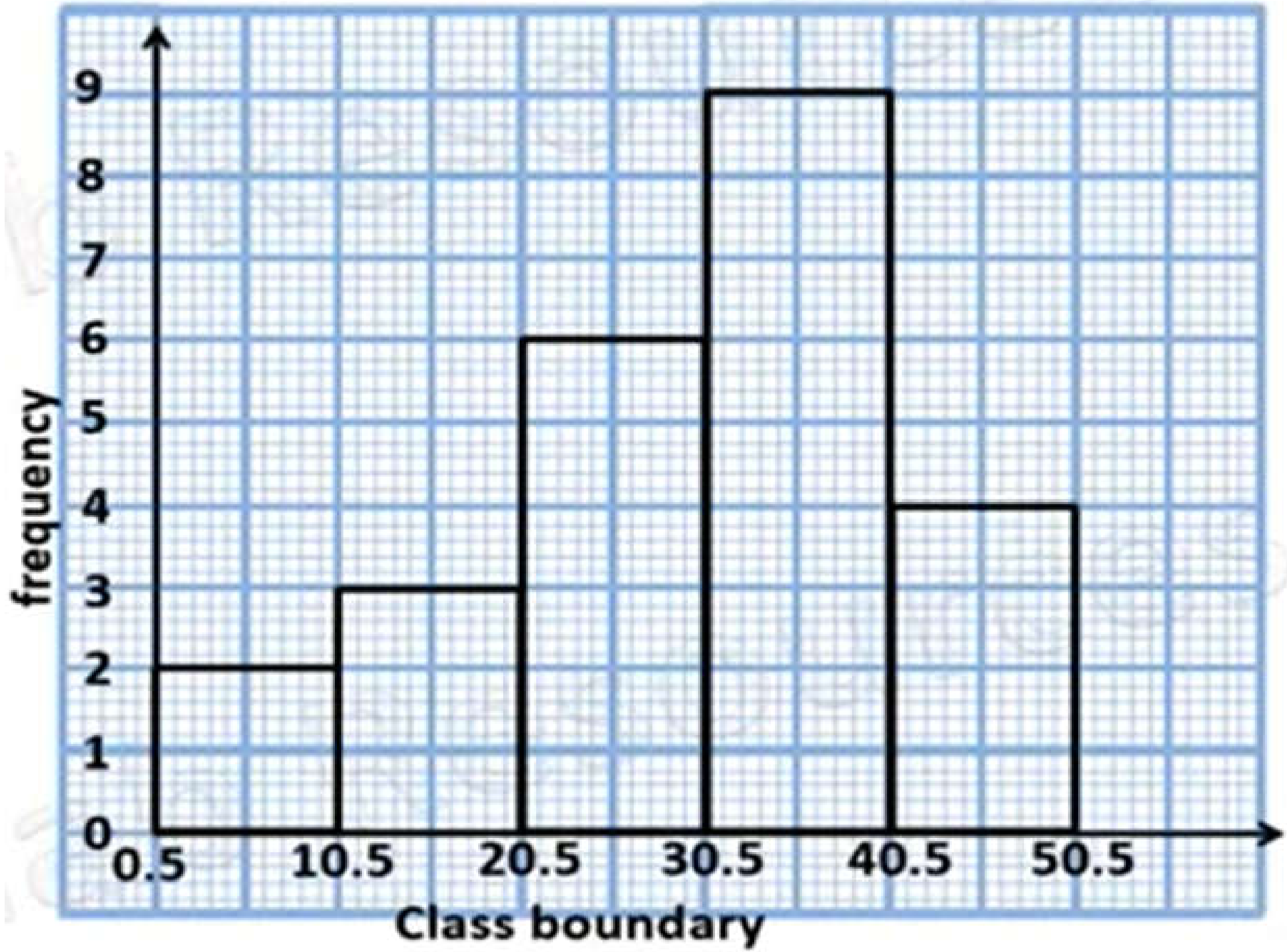
Problem 1

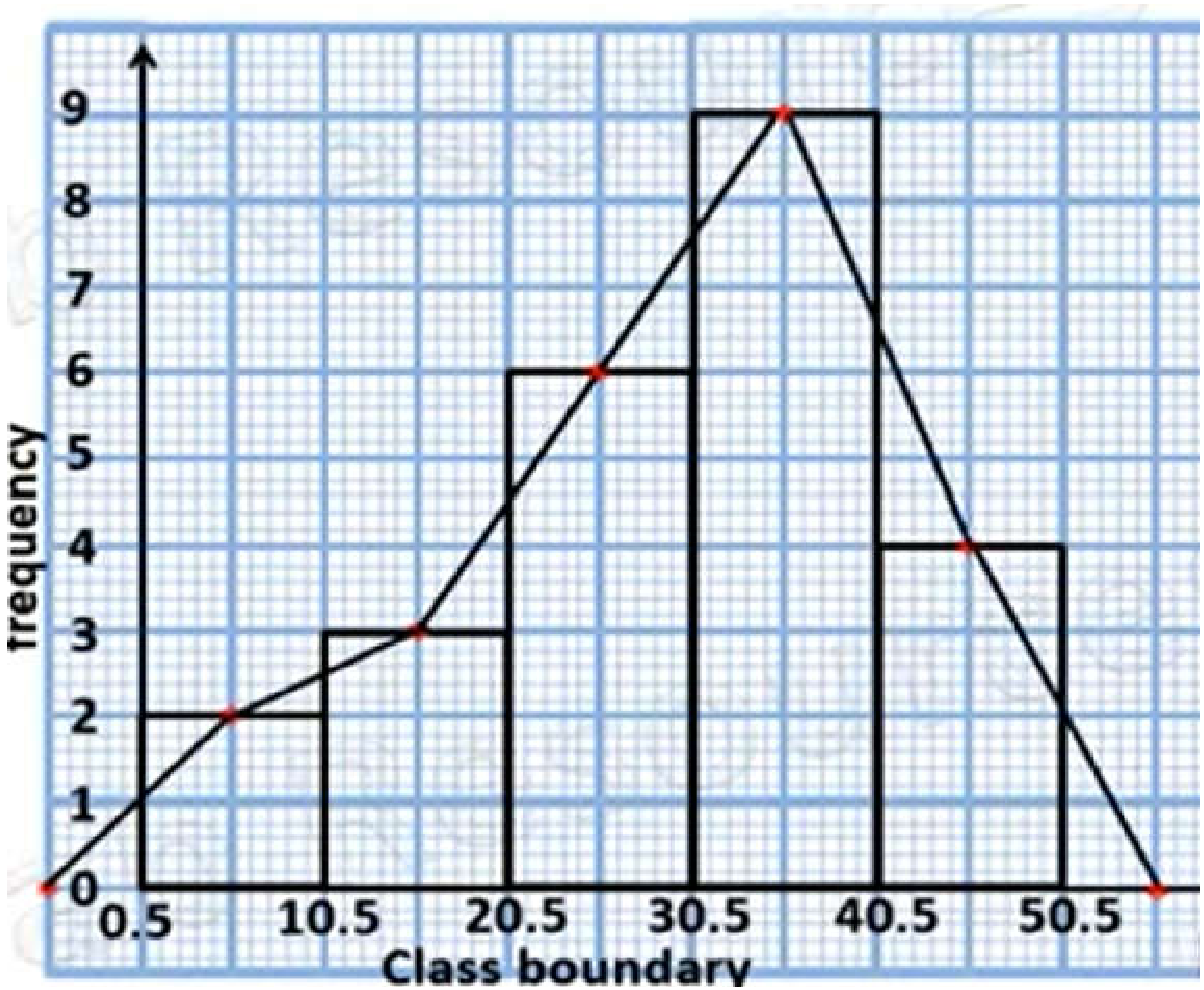
Use the frequency table below, draw the

(a) histogram (b) Frequency polygon of the distribution

Age	Frequency
1 – 10	2
11 – 20	3
21 – 30	6
31 – 40	9
41 – 50	4

Age	Frequency	CB
1 – 10	2	0.5 – 10.5
11 – 20	3	10.5 – 20.5
21 – 30	6	20.5 – 30.5
31 – 40	9	30.5 – 40.5
41 – 50	4	40.5 – 50.5


frequency
9
8
7
6
5
4
3
2
1
0
0.5
10.5
20.5
30.5
40.5
50.5
Class boundary



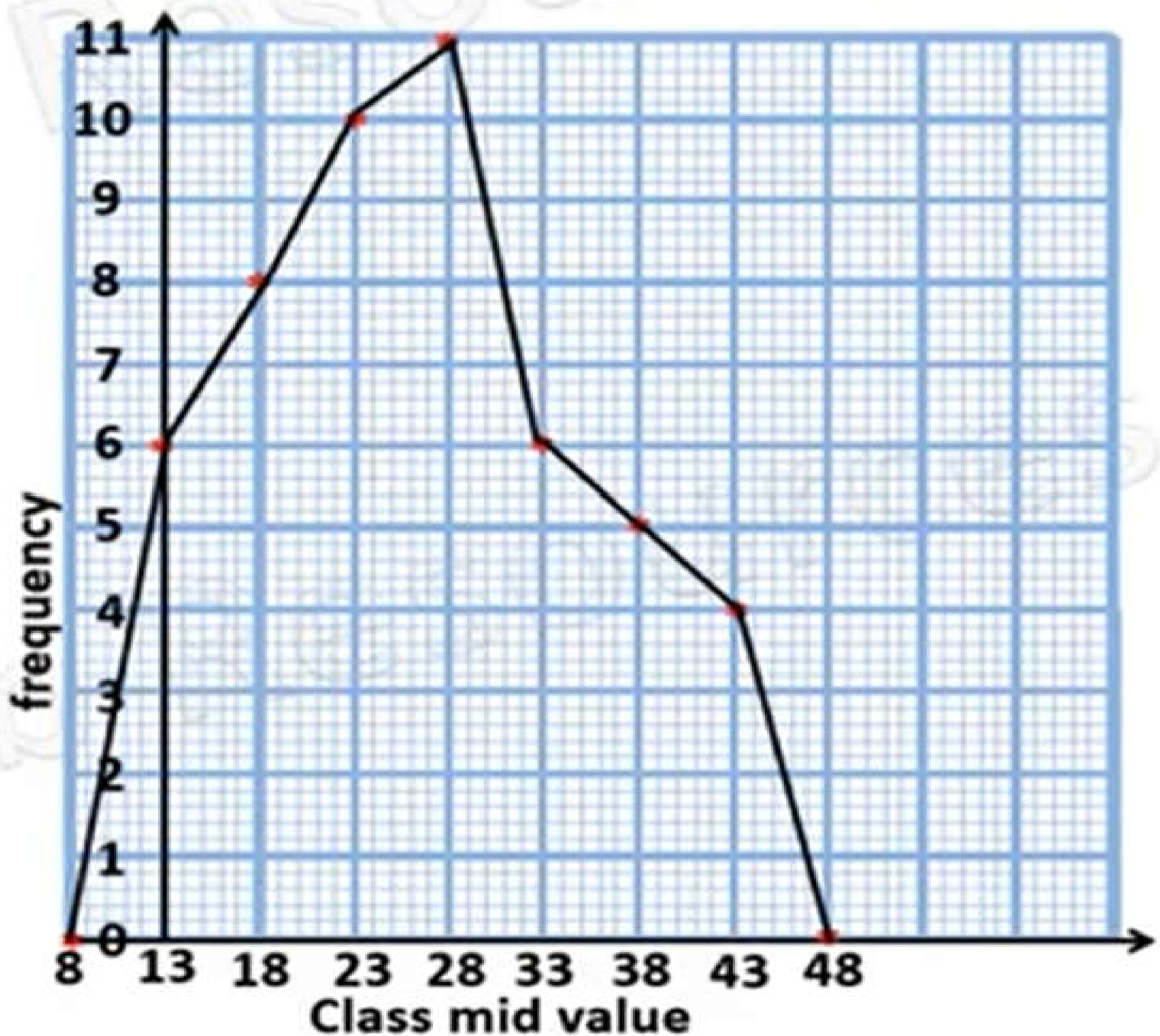
frequency
9
8
7
6
5
4
3
2
1
0
0.5
10.5
20.5
30.5
40.5
50.5
Class boundary

Problem 2

The ages of 50 people in a hamlet are given in the table below, draw the frequency polygon of the distribution

Age	Frequency
11 – 15	6
16– 20	8
21 – 25	10
26 – 30	11
31 – 35	6
36 – 40	5
41 – 45	4

Age	Frequency	CMV
11 – 15	6	13
16– 20	8	18
21 – 25	10	23
26 – 30	11	28
31 – 35	6	33
36 – 40	5	38
41 – 45	4	43



11
10
9
8
7
6
5
4
3
2
1
0
frequency
8
13
18
23
28
33
38
43
48
Class mid value

Example 1

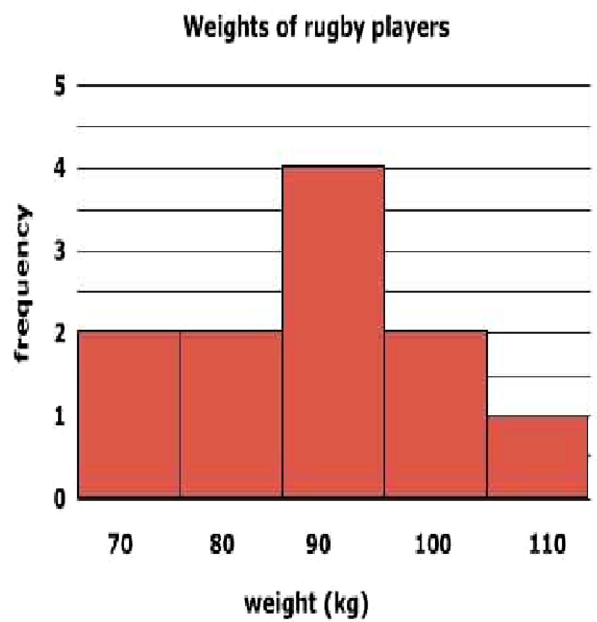
The weights of the rugby players on the university team are given below. How can this be presented as a histogram?

Weight (Kg)	67	72	78	81	86	89	90	93	97	98	113

Solution

The lowest value is 67 and highest is 113. As there isn't a lot of data, a smaller number of classes would be appropriate. Choosing 5 classes as shown gives class midpoints which are nice whole numbers.

Weight	Class midpoint	Frequency
65-75	70	2
75-85	80	2
85-95	90	4
95-105	100	2
105-115	110	1



Example 2

How would the weights of the rugby players in example 7 be presented as a frequency polygon?

Solution